CLASS 10 — SCIENCE · EASY LEVEL UNIT TEST SET 1

Chapters 9–12: **Light - Reflection and Refraction, The Human Eye and the Colourful World, Electricity, Magnetic Effects of Electric Current** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Light travels in 1

(a) zigzag paths **(b)** circles **(c)** straight lines **(d)** spirals

2. The eye is a natural 1

(a) musical instrument **(b)** optical instrument

(c) prism **(d)** heat engine

3. Electric current is the rate of flow of 1

(a) energy **(b)** mass **(c)** charge **(d)** heat

4. The region around a magnet in which its force can be felt is the 1

(a) electric field **(b)** light field

(c) gravitational field **(d)** magnetic field

5. The bending of light when it passes from one medium to another is 1

(a) reflection **(b)** diffraction **(c)** dispersion **(d)** refraction

6. A person who cannot see distant objects clearly is 1

(a) myopic **(b)** hypermetropic **(c)** colour blind **(d)** presbyopic

7. The resistance of a wire depends on 1

(a) its length **(b)** the time **(c)** the current only **(d)** the voltage only

8. A current-carrying wire produces 1

(a) only light **(b)** no field **(c)** only sound **(d)** a magnetic field

9. A lens is a transparent material bounded by 1

(a) no surface **(b)** four surfaces

(c) one plane surface only **(d)** two spherical surfaces

10. A prism has 1

(a) no flat surface

(b) four faces only

(c) two triangular bases and three rectangular lateral surfaces

(d) one curved surface

11. Two resistors of 5 ohm each in series have a total of 1

(a) 2.5 ohm **(b)** 25 ohm **(c)** 5 ohm **(d)** 10 ohm

12. The force on a current-carrying conductor is shown by 1

(a) Fleming's left-hand rule **(b)** Fleming's right-hand rule
(c) Joule's law **(d)** Snell's law

13. The bouncing back of light from a surface is called 1

(a) refraction **(b)** reflection **(c)** scattering **(d)** dispersion

14. The transparent bulge in front of the eye is the 1

(a) lens **(b)** cornea **(c)** iris **(d)** retina

15. Which of these is an insulator? 1

(a) Copper **(b)** Aluminium **(c)** Iron **(d)** Rubber

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. What is reflection of light? 2

17. Name the part of the eye where the image is formed. 2

18. Define electric current. 2

19. What is a magnetic field? 2

20. Which mirror gives a diminished image always? 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Name the type of mirror used for 3

(a) shaving
(b) torch
(c) rear-view of vehicle.

22. Name the parts of the eye which 3

(a) control the light entering
(b) focus the light
(c) receive the image.

23. Find the charge flowing in 5 s through a wire carrying a current of 2 A. 3

24. Draw (describe) the field lines of a bar magnet and mention the direction. 3

OR

Find the refractive index of diamond if light travels in it at 1.25×10^8 m/s. 3

25. What are the causes of presbyopia? 3

OR

List three factors on which the resistance of a wire depends. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Describe the images formed by a concave mirror when the object is 5

(a) at infinity
(b) at C
(c) between P and F.

OR

Describe the structure of the human eye and the functions of its parts. 5

27. State Ohm's law. Draw (describe) the V-I graph and explain what its slope represents. 5

OR

Describe the properties of magnetic field lines. 5

END OF QUESTION PAPER **Total: 50 Marks**

MASTERMIND